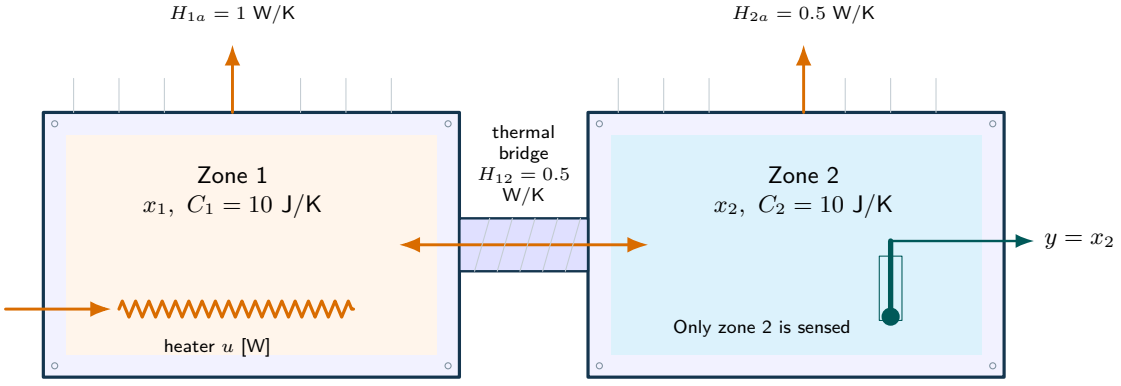

Temperatures are deviations from a fixed ambient temperature

Two energy stores

$C_1 = C_2 = 10$ J/K.
x_1, x_2: ambient-relative temperatures.
Input heat enters zone 1 only.

Coupled balances

$10\dot{x}_1 = u - 1.5x_1 + 0.5x_2$.
$10\dot{x}_2 = 0.5x_1 - x_2$.
All right-hand terms are heat rates.

Sensing and estimation

$y = x_2$ supplies one measured coordinate. An observer reconstructs both states from the applied input and this output.